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Rider; Tomer et al.

Facilitating dynamic detection and intelligent use of segmentation on flexible display screens

Abstract

A mechanism is described for facilitating dynamic detection and intelligent use of segmentation on flexible display screens according to one embodiment. A method of embodiments, as described herein, includes detecting, via one or more touch sensors, alterations in current in and around one or more areas of a flexible display screen, where the alterations represent pressure being applied to cause at least one of bending, rolling, and curving of the flexible display screen at the one or more areas. The method may further include dividing the flexible display screen into a plurality of zones corresponding to the one or more areas, where the marking/dividing logic is further to mark a plurality of portions of the plurality of zones to serve as a plurality of segments. The method may further include facilitating displaying of contents via the plurality of segments of the flexible display screen.

Inventors: Rider; Tomer (Naahryia, IL), Olmo; Ramon C. (Hillsboro, OR), Levy; Dor (Jerusalem, IL), Taite; Shahrar (Kfar Saba, IL)

Applicant: Intel Corporation (Santa Clara, CA)

Family ID: 1000008778591

Assignee: Intel Corporation (Santa Clara, CA)

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9727087	12/2016	Rider	N/A	N/A
10545536	12/2019	Rider	N/A	N/A
11216038	12/2021	Rider	N/A	N/A
11573607	12/2022	Rider	N/A	N/A
2010/0056223	12/2009	Choi et al.	N/A	N/A
2010/0164888	12/2009	Okumura	N/A	N/A
2012/0235894	12/2011	Phillips	N/A	N/A
2012/0306929	12/2011	Chalkov	N/A	N/A
2013/0176248	12/2012	Shin et al.	N/A	N/A
2013/0285922	12/2012	Alberth, Jr.	N/A	N/A
2013/0314349	12/2012	Chien	N/A	N/A
2014/0035869	12/2013	Yun et al.	N/A	N/A
2014/0049464	12/2013	Kwak	N/A	N/A
2014/0240264	12/2013	Im et al.	N/A	N/A
2014/0306908	12/2013	Nagaraju	N/A	N/A
2015/0185781	12/2014	Yu	N/A	N/A
2015/0261254	12/2014	Hiroki	N/A	N/A
2015/0296062	12/2014	Lee	N/A	N/A
2016/0109973	12/2015	Kim et al.	N/A	N/A
2022/0212096	12/2021	Delaporte	N/A	G06F 1/1677
2024/0069601	12/2023	Kim	N/A	G06F 1/1652

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2546720	12/2012	EP	N/A
2546720	12/2015	EP	N/A
WO-0153919	12/2000	WO	G06F 1/1616
WO-2021104008	12/2020	WO	G06F 1/1618
WO-2022260457	12/2021	WO	G06F 1/1652

OTHER PUBLICATIONS

Patent Cooperation Treaty, "International Search Report," issued in connection with International Patent Application No. PCT/US2016/017182, dated May 13, 2016, 3 pages. cited by applicant
International Searching Authority, "Written Opinion of the International Searching Authority," issued in connection with International Patent Application No. PCT/US2016/017182, dated May 13, 2016, 8 pages. cited by applicant

United States Patent and Trademark Office, "Non-Final Action," issued in connection with U.S. Appl. No. 14/668,664 on Jun. 3, 2016, 9 pages. cited by applicant

United States Patent and Trademark Office, "Final Action," issued in connection with U.S. Appl. No. 14/668,664 on Dec. 9, 2016, 10 pages. cited by applicant

United States Patent and Trademark Office, "Notice of Allowance," issued in connection with U.S. Appl. No. 14/668,664 on Apr. 21, 2017, 9 pages. cited by applicant

Patent Cooperation Treaty, "International Preliminary Report on Patentability," issued in connection with International Patent Application No. PCT/US2016/017182, dated Oct. 5, 2017, 10 pages. cited by applicant

United States Patent and Trademark Office, "Non-Final Action," issued in connection with U.S. Appl. No. 15/561,453 on Aug. 16, 2018, 11 pages. cited by applicant

United States Patent and Trademark Office, "Final Action," issued in connection with U.S. Appl. No. 15/561,453 on Jun. 3, 2019, 10 pages. cited by applicant

United States Patent and Trademark Office, "Notice of Allowance," issued in connection with U.S. Appl. No. 15/561,453 on Sep. 19, 2019, 8 pages. cited by applicant

United States Patent and Trademark Office, "Non-Final Action," issued in connection with U.S. Appl. No. 16/773,648 on Sep. 1, 2020, 7 pages. cited by applicant

United States Patent and Trademark Office, "Final Action," issued in connection with U.S. Appl. No. 16/773,648 on May 6, 2021, 10 pages. cited by applicant

United States Patent and Trademark Office, "Notice of Allowance," issued in connection with U.S. Appl. No. 16/773,648 on Aug. 27, 2021, 8 pages. cited by applicant

United States Patent and Trademark Office, "Notice of Allowance," issued in connection with U.S. Appl. No. 16/773,648 on Sep. 10, 2021, 2 pages. cited by applicant

United States Patent and Trademark Office, "Non-Final Action," issued in connection with U.S. Appl. No. 17/537,346 on Jun. 16, 2022, 7 pages. cited by applicant

United States Patent and Trademark Office, "Notice of Allowance," issued in connection with U.S. Appl. No. 17/537,346 on Sep. 30, 2022, 8 pages. cited by applicant

United States Patent and Trademark Office, "Non-Final Office Action," issued in connection with U.S. Appl. No. 18/149,033, dated Apr. 26, 2023, 13 pages. cited by applicant

United States Patent and Trademark Office, "Final Office Action," issued in connection with U.S. Appl. No. 18/149,033, dated Nov. 17, 2023, 14 pages. cited by applicant

Background/Summary

RELATED APPLICATIONS (1) This patent is a continuation of U.S. patent application Ser. No. 18/149,003, filed Dec. 30, 2022, which is a continuation of U.S. patent application Ser. No. 17/537,346, filed Nov. 29, 2021, and granted as U.S. Pat. No. 11,573,607, on Feb. 7, 2023, which is a continuation of U.S. patent application Ser. No. 16/773,648, filed Jan. 27, 2020, and granted as U.S. Pat. No. 11,216,038, on Jan. 4, 2022, which is a continuation of U.S. patent application Ser. No. 15/561,453, filed Sep. 25, 2017, and granted as U.S. Pat. No. 10,545,536, on Jan. 28, 2020, which is a § 371 National Stage application of PCT Application Serial No. PCT/US16/17182, filed Feb. 9, 2016, which is a continuation of U.S. patent application Ser. No. 14/668,664, filed Mar. 25, 2015, and granted as U.S. Pat. No. 9,727,087 on Aug. 8, 2017. U.S. patent application Ser. No. 17/537,346, U.S. Pat. Nos. 11,216,038, 10,545,536, PCT Application Serial No. PCT/US16/17182 and U.S. Pat. No. 9,727,087 are hereby incorporated by reference in their entireties.

FIELD

(1) Embodiments described herein generally relate to computers. More particularly, embodiments relate to facilitating dynamic detection and intelligent use of segmentation on flexible display screens.

BACKGROUND

(2) With the growth in computing technology, display screens, including flexible display screens, are also gaining popularity and noticeable traction in becoming a mainstream technology as seen being employed in various devices, such as televisions, wearable devices, smartphones, tablet computers, etc., and even as standalone flexible displays. However, conventional techniques treat flexible displays as single displays and are severely limited in their application.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Embodiments are illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings in which like reference numerals refer to similar elements.
- (2) FIG. 1 illustrates a computing device employing a dynamic segmentation mechanism according to one embodiment.
- (3) FIG. 2A illustrates a dynamic segmentation mechanism according to one embodiment.
- (4) FIG. 2B illustrates a flexible display screen according to one embodiment.
- (5) FIG. 2C illustrates a flexible display screen according to one embodiment.
- (6) FIG. 2D illustrates a flexible display screen according to one embodiment.
- (7) FIG. 2E illustrates a flexible display screen according to one embodiment.
- (8) FIG. 3 illustrates a method for facilitating dynamic segmentation and intelligent use of segmentation on flexible display screens according to one embodiment.
- (9) FIG. 4 illustrates computer system suitable for implementing embodiments of the present disclosure according to one embodiment.
- (10) FIG. 5 illustrates computer environment suitable for implementing embodiments of the present disclosure according to one embodiment.

DETAILED DESCRIPTION

(11) In the following description, numerous specific details are set forth. However, embodiments, as described herein, may be practiced without these specific details. In other instances, well-known circuits, structures and techniques have not been shown in details in order not to obscure the understanding of this description.

(12) Embodiments provide for proactively identifying various curves and bends on a flexible display screen (also referred to as “flexible screen” or “flexible display”) to dynamically segment the flexible screen into multiple areas with each area serving as a screen, enabling multiple new user interfaces and experiences on the single flexible screen in its different states. In one embodiment, a single flexible screen may be used as having different display areas providing different contents by proactively detecting and using the flexible screen's various curves and bends.

(13) FIG. 1 illustrates a computing device **100** employing a dynamic segmentation mechanism **110** according to one embodiment. Computing device **100** servers a host machine for hosting dynamic segmentation mechanism (“segmentation mechanism”) **110** that may include any number and type of components, as illustrated in FIG. 2, to facilitate detection and use of segmentations in flexible display screens as will be further described throughout this document.

(14) Computing device **100** may include any number and type of communication devices, such as large computing systems, such as server computers, desktop computers, etc., and may further include set-top boxes (e.g., Internet-based cable television set-top boxes, etc.), global positioning system (GPS)-based devices, etc. Computing device **100** may include mobile computing devices serving as communication devices, such as cellular phones including smartphones, personal digital assistants (PDAs), tablet computers, laptop computers (e.g., Ultrabook™ system, etc.), e-readers, media internet devices (MIDs), media players, smart televisions, television platforms, intelligent devices, computing dust, media players, smart windshields, smart windows, head-mounted displays (HMDs) (e.g., optical head-mounted display (e.g., wearable glasses, head-mounted binoculars, gaming displays, military headwear, etc.), and other wearable devices (e.g., smartwatches, bracelets, smartcards, jewelry, clothing items, etc.), etc.

(15) It is contemplated and to be noted that embodiments are not limited to computing device **100** and that embodiments may be applied to and used with any form or type glass that is used for viewing purposes, such as smart windshields, smart windows (e.g., smart window by Samsung®, etc.), and/or the like. Similarly, it is contemplated and to be noted that embodiments are not limited to any particular type of computing device and that embodiments may be applied and used with any number and type of computing devices; however, throughout this document, the focus of the discussion may remain on wearable devices, such as wearable glasses, etc., which are used as examples for brevity, clarity, and ease of understanding.

(16) In some embodiments, computing device **100** may include a large(r) computing system (e.g., server computer, desktop computer, laptop computer, etc.), such that a flexible display screen may be part of this large(r) computing system where the flexible display screen may be a part or an extension screen of a main display screen, where the main screen itself may be flexible or static.

(17) Computing device **100** may include an operating system (OS) **106** serving as an interface between hardware and/or physical resources of the computer device **100** and a user. Computing device **100** further includes one or more processors **102**, memory devices **104**, network devices, drivers, or the like, as well as input/output (I/O) sources **108**, such as one or more touchable and/or non-touchable flexible display screen(s) (e.g., rolling screens, bendable screens, curving screens, etc.), touchscreens, touch panels, touch pads, virtual or regular keyboards, virtual or regular mice, etc. A touchable and/or non-touchable flexible display screen may be the same as or similar to flexible display screens **230A**, **230B**, **230C**, **230D**, and **230E** of FIGS. 2A, 2B, 2C, 2D and 2E, respectively.

(18) It is to be noted that terms like “node”, “computing node”, “server”, “server device”, “cloud computer”, “cloud server”, “cloud server computer”, “machine”, “host machine”, “device”, “computing device”, “computer”, “computing system”, and the like, may be used interchangeably

throughout this document. It is to be further noted that terms like “application”, “software application”, “program”, “software program”, “package”, “software package”, “code”, “software code”, and the like, may be used interchangeably throughout this document. Also, terms like “job”, “input”, “request”, “message”, and the like, may be used interchangeably throughout this document. It is contemplated that the term “user” may refer to an individual or a group of individuals using or having access to computing device **100**.

(19) FIG. 2A illustrates a dynamic segmentation mechanism **110** according to one embodiment. In one embodiment, segmentation mechanism **110** may include any number and type of components, such as (without limitation): detection/appropriation engine **201** including touch segment detection logic (“touch logic”) **203**, non-touch segment detection logic (“non-touch logic”) **205**, and marking/dividing logic **207**; application/use engine **211** including activation/user interface logic **213** and content/use logic **215**; preferences logic **217**; and communication/compatibility logic **219**.

(20) Computing device **100** (e.g., handheld device, wearable device, smart window, etc.) may further include any number and type of other components, such as capturing/sensing components **221** (e.g., capacitor touch sensors (“touch sensors”) **231**, current delta non-touch sensors (“non-touch sensors”) **233** (e.g., delta-sigma modulator, etc.), cameras, microphones, etc.), output components **223** (e.g., touch/non-touch flexible display screen **230A**, such as rolling screen, bendable screen, curving screen, etc.), etc.

(21) It is contemplated that flexible screen **230A** may not be part of computing device **100** and that it may be a standalone display screen and may be in communication with computing device **100**. For example and in one embodiment, computing device **100** may be a smart window or a handheld device having flexible display screen **235** that may include one or more of a rolling screen that is capable of being rolled in or more ways, bendable screen that is capable of being bent in one or more ways, curving screens that can be curved in one or more ways, etc., and further, flexible display screen **230A** may be a touch screen or a non-touch screen.

(22) As aforementioned with reference to FIG. 1, in some embodiments, computing device **100** may include a large(r) computing system (e.g., server computer, desktop computer, laptop computer, etc.), such that flexible display screen **230A** may be part of this large(r) computing system where flexible display screen **230A** may be a part or an extension screen of a main display screen, where the main screen itself may be flexible or static.

(23) Further, for example and in one embodiment, capturing/sensing components **221** may include any number and type of components, such as touch sensors **231**, non-touch sensors **233**, cameras, microphones, etc., for performing detection and sensing tasks for segmentation of flexible screen **230A** of output components **223** as will be further described below.

(24) Capturing/sensing components **221** may further include any number and type of capturing/sensing devices, such as one or more sending and/or capturing devices (e.g., cameras, microphones, biometric sensors, chemical detectors, signal detectors, wave detectors, force sensors (e.g., accelerometers), illuminators, etc.) that may be used for capturing any amount and type of visual data, such as images (e.g., photos, videos, movies, audio/video streams, etc.), and non-visual data, such as audio streams (e.g., sound, noise, vibration, ultrasound, etc.), radio waves (e.g., wireless signals, such as wireless signals having data, metadata, signs, etc.), chemical changes or properties (e.g., humidity, body temperature, etc.), biometric readings (e.g., figure prints, etc.), environmental/weather conditions, maps, etc. It is contemplated that “sensor” and “detector” may be referenced interchangeably throughout this document. It is further contemplated that one or more capturing/sensing components **221** may further include one or more supporting or supplemental devices for capturing and/or sensing of data, such as illuminators (e.g., infrared (IR) illuminator), light fixtures, generators, sound blockers, etc.

(25) It is further contemplated that in one embodiment, capturing/sensing components **221** may further include any number and type of sensing devices or sensors (e.g., linear accelerometer) for sensing or detecting any number and type of contexts (e.g., estimating horizon, linear acceleration,

etc., relating to a mobile computing device, etc.). For example, capturing/sensing components **221** may include any number and type of sensors, such as (without limitations): accelerometers (e.g., linear accelerometer to measure linear acceleration, etc.); inertial devices (e.g., inertial accelerometers, inertial gyroscopes, micro-electro-mechanical systems (MEMS) gyroscopes, inertial navigators, etc.); gravity gradiometers to study and measure variations in gravitation acceleration due to gravity, etc.

(26) For example, capturing/sensing components **221** may further include (without limitations): audio/visual devices (e.g., cameras, microphones, speakers, etc.); context-aware sensors (e.g., temperature sensors, facial expression and feature measurement sensors working with one or more cameras of audio/visual devices, environment sensors (such as to sense background colors, lights, etc.), biometric sensors (such as to detect fingerprints, etc.), calendar maintenance and reading device), etc.; global positioning system (GPS) sensors; resource requestor; and trusted execution environment (TEE) logic. TEE logic may be employed separately or be part of resource requestor and/or an I/O subsystem, etc.

(27) Computing device **100** may further include one or more output components **223** to remain in communication with one or more capturing/sensing components **221** and one or more components of segmentation mechanism **110** to facilitate displaying of images, playing or visualization of sounds, displaying visualization of fingerprints, presenting visualization of touch, smell, and/or other sense-related experiences, etc. For example and in one embodiment, output components **223** may include (without limitation) one or more of light sources, display devices or screens, audio speakers, bone conducting speakers, olfactory or smell visual and/or non/visual presentation devices, haptic or touch visual and/or non-visual presentation devices, animation display devices, biometric display devices, X-ray display devices, etc.

(28) Computing device **100** may be in communication with one or more repositories or databases over one or more networks, where any amount and type of data (e.g., real-time data, historical contents, metadata, resources, policies, criteria, rules and regulations, upgrades, etc.) may be stored and maintained. Similarly, computing device **100** may be in communication with any number and type of other computing devices, such as HMDs, wearable devices, smart windows, mobile computers (e.g., smartphone, a tablet computer, etc.), desktop computers, laptop computers, etc., over one or more networks (e.g., cloud network, the Internet, intranet, Internet of Things (“IoT”), proximity network, Bluetooth, etc.).

(29) It is contemplated that computing device **100** may include one or more software applications (e.g., device applications, hardware components applications, business/social application, websites, etc.) in communication with segmentation mechanism **110**, where a software application may offer one or more user interfaces (e.g., web user interface (WUI), graphical user interface (GUI), touchscreen, etc.) to work with and/or facilitate one or more operations or functionalities of segmentation mechanism **110**.

(30) In one embodiment, computing device **100** may include a flexible display screen-based device, such as a handheld device, a wearable device, a smart windows, laptop computer, desktop computer, etc., having at least one flexible display screen which may be touchable or non-touchable. Further, flexible display screen **230A** may be of any size, such as a micro-screen mounted on a smartcard or a smart bracelet to a very large screen that is wall-mounted or billboard-mounted, etc., based on any number and type of techniques or technologies, such as (without limitation) electrochromic, photochromic, thermochromic, or suspended particles, etc. It is contemplated and to be noted that embodiments are not limited to any particular number and type of flexible screen **230A** being standalone or device-based, small or large, single layered or block of layers, or depending on any particular type or form of technology, etc.

(31) In one embodiment, as illustrated, segmentation mechanism **110** includes detection/appropriation engine **201** for facilitated sensing and detection of one or more segments of flexible screen **230A** by identifying any number of curves and bends using one or more

components, such as touch logic **203**, non-touch logic **205**, marking/dividing logic **207**, touch sensor **231**, and non-touch sensor **233**.

(32) In one embodiment, in case of flexible screen **230A** being a touch-based screen, touch logic **203** may be used to facilitate touch sensor **231** to detect any changes in the running charge of flexible screen **230A** at an axis when flexible screen is bent (rolled or curved) at the axis, because when flexible screen **230A** is bent at a certain axis, the charge around that axis is altered. For example, under normal circumstances, such as when flexible screen **230A** remains unbent, the polarity charge of flexible screen **230A** continues to run in constant current streams until flexible screen **230A** is bent at an axis which can lead to changes in pixel proximity around the axis area which further leads to differences or modifications in the current around that axis area. In one embodiment, as aforementioned, touch logic **203** facilitates touch sensor **231** to detect and identify such changes in the current or charges around the axis area of flexible screen **230A**.

(33) In another embodiment, in case of flexible screen **230A** being a non-touch screen, non-touch logic **205** may be used to facilitate non-touch sensor **233** to track and extract any indication of flexible screen **230A** being bent (rolled or curved) by measuring small current changes over a period of time in a specific area of flexible screen **230A**, where specific area includes an axis area at which flexible screen **230A** is bent. For example, the change in the current may indicate screen bending of flexible screen **230A** around an axis by measuring charge differences on the bent axis as facilitated by non-touch logic **205** using non-touch sensor **233**.

(34) It is contemplated that embodiments are not merely limited to “bending” and that flexible screen **230A** may be “rolled” or “curved, as desired or necessitated, and that “bending” is mentioned throughout this document as an example for brevity, clarity, and ease of understanding.

(35) Once any bends, rolls, and/or curves on flexible screen **230A** are detected and measured by touch logic **203** and non-touch logic **205** using touch sensor **231** and non-sensor **233**, respectively, any data or information obtained from these detections and measurements is forwarded on to marking/dividing logic **207** for further processing. For example, touch logic **203**, via touch sensor **231**, may detect and measure any changes in the charges around one or more axis areas due to changes in screen pixel proximity in those axis areas which is caused by bending of touch-based flexible screen **230A**. This measurement data is then forwarded on to marking/dividing logic **207** to be used to divide flexible screen **230A** at locations corresponding to the identified axis areas as multiple zones, where these zones are then marked as parts or segments to then be used as separate display screens for displaying different contents on flexible screen **230A**.

(36) Similarly, for example, non-touch logic **205**, via non-touch sensor **233**, may detect and measure any differences or changes in the current charge, over time, around specific areas. This measurement technique includes using non-touch sensor **233** for extraction of small changes in current charges as detected in one or more specific areas over a period of time and continuously measuring any differences detected between previous charges and current charges to identify and regard the one or more specific areas as bend areas or axis areas. This measure of axis areas is used by marking/dividing logic **207** to divide flexible screen **230A** at locations corresponding to the identified axis areas as multiple zones, where these zones are then marked as parts or segments to then be used as separate display screens for displaying different contents on flexible screen **230A**.

(37) In one embodiment, activation/user interface logic **213** of application/use engine **211** may be used to activate the divided and marked segment activating these segments as displays and assigning them their user interfaces. As further illustrated with respect to FIGS. 2B-2E, each segment of flexible screen **230A** may be used as a separate display screen capable of providing content that may be distinct and different from the contents provided through other segments of flexible screen **230A**. For example, if flexible screen **230A** is bent and divided into two segments, one of the two segments may display a website showing local weather details, while the other segment may show a video relating to the local weather or something entirely different, such as a sports website, a television news channel, a movie, etc., or it may simply be left blank or turned off.

(38) In one embodiment, activation/user interface logic **213** activates each segment to enable it to display content and further, activation/user interface logic **213** assigns a separate user interface to each segment to allow it to play content that may be distinguished from contents of other segments on the same flexible screen **230A**. Moreover, in one embodiment, content/use logic **215** may be used to facilitate each segment to provide it content through its assigned user interface. For example, upon having the segments activated and assigned their corresponding interfaces by activation/user interface **213**, each segment may then be facilitated to accept any amount and type of content and with the ability to display the content as facilitated by content/use logic **215**.

(39) As aforementioned, it is contemplated and to be noted that although “bending” of flexible screen **230A** is used throughout this document, embodiments are not limited as such and that forms of pressure, such as “curving”, “rolling”, etc., of flexible screen **230A** may also be applied and performed for achieving segmentation of flexible screen **230A** as facilitated by segmentation mechanism **110** and as described above.

(40) In one embodiment, segmentation mechanism **110** further includes preference logic **217** to allow a user of computing device **100** to set their own preferences on how they wish to use the multiple segments of flexible screen **230A**. For example, in one embodiment, the user may choose to predefine or preset the different types or categories of contents they wish to have displayed on different segments of flexible screen **110**. For example, a user who is an investor or works in the finance industry may wish to have the stock market numbers displayed at all time on the right segment or curve of flexible screen **230A**, and/or the like. Similarly, the user may wish to have family photos along with current time and weather displayed at all time on the left segment or curve of flexible screen **230A**, and/or the like. In some embodiment, users may wish to have all segments display a single content, such as a movie, etc., such as having portions of a single movie screen collectively displayed using multiple segments (e.g., cut the movie screen or a picture or a website content in six portions to be displayed by six segments of flexible screen **230A**).

(41) In one embodiment, the user may access a user interface on computing device **100** to set any type of preferences or personal settings that may then be taken into consideration and applied by segmentation mechanism **110**. Further, it is contemplated that embodiments are not limited to any of the preferences described above and that users may choose to set and reset any number and type of personal settings, as desired or necessitated.

(42) In one embodiment and for example, content/use logic **215**, in working with activation/user interface logic **213** and preferences logic **217**, allows for interaction and communication between two or more segments, allowing the user to efficiently perform multiple tasks (referred to as “multitasking”) based on user preferences. Similarly, in case of computing device **100** being a smartphone with bending abilities, computing device **100** and flexible screen **112** may be bent such that content/use logic **215**, based on the user's preference settings, may allow for dividing different widgets on each segment of multiple segments of flexible screen **230A**.

(43) Further, as illustrated with reference to FIG. 2E, segmentation of flexible screen **230A** may further allow for partitioning of flexible screen **230A** into different segments providing additional screens which may be extremely valuable in certain activities, such as gaming. For example, using user preferences as determined by preference logic **217** and as facilitated by content/use logic **215**, in case of a war game, one segment may display the game and its progression, while another segment may display weapons menu to efficiently and easily control and play the game.

(44) In one embodiment, in performing proactive detection of bending, rolling, curving, etc., of flexible screen **230A** using segmentation logic **110**, any premature or accidental bending, rolling, curving, etc., of flexible screen **230A** or computing device **100** may be supervised, timely detected, and communicated, via communication/compatibility logic **219**, to the user via user interface at computing device **100**. This way the user may be warned and computing device **100** may be protected. In one embodiment, as aforementioned, flexible screen **230A** may be curved or rolled along with being bent and as such, flexible screen **230A** and its segments may be used to zoom-in

and zoom-out of content by, for example, dynamically alternatively curving flexible screen **230A** between convex state and concave state. It is contemplated that embodiments are not merely limited to any of the examples discussed above, but that any number and type of settings, triggering events, etc., may be set or reset by user, via preferences logic **217**, to correlate with a display event, such as increasing or decreasing volume on a screen flex event.

(45) Communication/compatibility logic **219** may be used to facilitate dynamic communication and compatibility between computing device **100** and any number and type of other computing devices (such as wearable computing devices, mobile computing devices, desktop computers, server computing devices, etc.), processing devices (e.g., central processing unit (CPU), graphics processing unit (GPU), etc.), capturing/sensing components **221** (e.g., capacitor touch sensors, current delta sensors, non-visual data sensors/detectors, such as audio sensors, olfactory sensors, haptic sensors, signal sensors, vibration sensors, chemicals detectors, radio wave detectors, force sensors, weather/temperature sensors, body/biometric sensors, scanners, etc., and visual data sensors/detectors, such as cameras, etc.), user/context-awareness components and/or identification/verification sensors/devices (such as biometric sensors/detectors, scanners, etc.), memory or storage devices, databases and/or data sources (such as data storage devices, hard drives, solid-state drives, hard disks, memory cards or devices, memory circuits, etc.), networks (e.g., cloud network, the Internet, intranet, cellular network, proximity networks, such as Bluetooth, Bluetooth low energy (BLE), Bluetooth Smart, Wi-Fi proximity, Radio Frequency Identification (RFID), Near Field Communication (NFC), Body Area Network (BAN), etc.), wireless or wired communications and relevant protocols (e.g., Wi-Fi®, WiMAX, Ethernet, etc.), connectivity and location management techniques, software applications/websites, (e.g., social and/or business networking websites, business applications, games and other entertainment applications, etc.), programming languages, etc., while ensuring compatibility with changing technologies, parameters, protocols, standards, etc.

(46) Throughout this document, terms like “logic”, “component”, “module”, “framework”, “engine”, “tool”, and the like, may be referenced interchangeably and include, by way of example, software, hardware, and/or any combination of software and hardware, such as firmware. Further, any use of a particular brand, word, term, phrase, name, and/or acronym, such as “flexible display screen”, “flexible screen”, “segmentation”, “segment”, “zone”, “bend”, “roll”, “curve”, “touch”, “non-touch”, “smart glass”, “wearable device”, etc., should not be read to limit embodiments to software or devices that carry that label in products or in literature external to this document.

(47) It is contemplated that any number and type of components may be added to and/or removed from segmentation mechanism **110** to facilitate various embodiments including adding, removing, and/or enhancing certain features. For brevity, clarity, and ease of understanding of segmentation mechanism **110**, many of the standard and/or known components, such as those of a computing device, are not shown or discussed here. It is contemplated that embodiments, as described herein, are not limited to any particular technology, topology, system, architecture, and/or standard and are dynamic enough to adopt and adapt to any future changes.

(48) FIG. 2B illustrates a flexible display screen **230B** according to one embodiment. As an initial matter, for the sake of brevity, clarity, and ease of understanding, many of the processes and components discussed above with respect to FIGS. 1-2A may not be discussed or repeated hereafter, such as with respect to FIGS. 2B-5. Further, it is contemplated and to be noted that flexible display screens **230B**, **230C**, **230D**, and **230E** of FIGS. 2B, 2C, 2D, and 2E, respectively, may be the same as or similar to flexible display screen **230A** of FIG. 2A and that this contemplation may not be repeated hereafter.

(49) The illustrated embodiment of flexible screen **230B** includes a rolling or rollable flexible screen (e.g., smart bracelet, smartwatch, etc.) that is shown in two different views, such as flat view **231** and rolled view **233**. In flat view **231**, flexible screen **230B** is shown in its basic or original state which, as shown in rolled view **233**, is then rolled into a circular shape like a watch or a

bracelet, etc. As previously described with reference to FIG. 2A, various bending, curving, and rolling areas on flexible screen **230B** may be detected and measured to lead to identification of multiple viewing zones **235** that are then marked as segments **237** to serve as multiple display screens (showing any amount and type of contents) using a single flexible screen **230B**. As aforementioned with respect to FIG. 2A, these multiple segments **237** may be used in any number of ways, such as serving as interactive display screens to allow for multitasking, gaming, etc. (50) FIG. 2C illustrates a flexible display screen **230C** according to one embodiment. The illustrated embodiment of flexible screen **230C** includes a curving or curve-able flexible screen (e.g., smart bangles, curving strips, etc.) that is shown in a curved state that may be obtained from pressuring at both ends of flexible screen **230C**. In curving flexible screen **230C**, a couple of viewing zones **245** and their corresponding segments **247** on flexible screen **230C** are obtained and used as previously discussed with reference to FIG. 2A.

(51) FIG. 2D illustrates a flexible display screen **230D** according to one embodiment. The illustrated embodiment of flexible screen **230D** includes another rolling or rollable flexible screen that is shown in a wavy state that may be obtained from pressuring at various parts of flexible screen **230D**. In waving flexible screen **230D**, multiple viewing zones **245** of various sizes (e.g., zone 1 is smaller than zones 2-4) and their corresponding segments **247** on flexible screen **230C** are obtained and used as previously discussed with reference to FIG. 2A.

(52) FIG. 2E illustrates a flexible display screen **230E** according to one embodiment. The illustrated embodiment of flexible screen **230E** includes a bending or bendable flexible screen that is shown in a bended state that may be obtained from pressuring in the middle and/or the two ends of flexible screen **230E**. As illustrated, in bending flexible screen **230E**, two segments **267A-B** are obtained as previously discussed with reference to FIG. 2A. In one embodiment, as illustrated, segments **267A-B** are used for gaming purposes, where segment **267A** is used as a weapons screen, while segment **267B** is used as a gameplay screen.

(53) FIG. 3 illustrates a method **300** for facilitating dynamic segmentation and intelligent use of segmentation on flexible display screens according to one embodiment. Method **300** may be performed by processing logic that may comprise hardware (e.g., circuitry, dedicated logic, programmable logic, etc.), software (such as instructions run on a processing device), or a combination thereof. In one embodiment, method **300** may be performed by segmentation mechanism **110** of FIGS. 1-2A. The processes of method **300** are illustrated in linear sequences for brevity and clarity in presentation; however, it is contemplated that any number of them can be performed in parallel, asynchronously, or in different orders. For brevity, many of the details discussed with reference to FIGS. 1 and 2A-E may not be discussed or repeated hereafter.

(54) Method **300** begins at block **301** with detecting pressure on a flexible display screen, where the pressure leads to bending, rolling, and/or curving of the flexible display screen. At block **303**, one or more areas of pressure are detected and, at block **305**, changes in current charges at the one or more axis areas are measured. As described with reference to FIG. 2A, in one embodiment, in case of the flexible screen being a touch screen, touch logic **203** may be used to facilitate one or more touch sensor(s) **231** (e.g., touch capacitor sensors) detect and identify any changes in the current charge around the one or more axis areas where the bending, rolling, and/or curving of the flexible screen takes place, such as when the pixel proximity of the flexible screen changes around these one or more axis areas due to at least one of bending, rolling, and/or curving of the flexible screen. In one embodiment, touch logic **203** may further facilitate the one or more touch sensor(s) **231** (e.g., touch capacitor sensors) of FIG. 2A to measure these changes or differences in the current charges around the one or more axis areas to, for example, determine capacitance or change in capacitance of the one or more axis areas.

(55) Similarly, as further described with reference to FIG. 2A, non-touch logic **205** may be used to facilitate one or more non-touch sensor(s) **233** (e.g., current delta sensors) to detect and extract current changes in and around one or more specific areas (e.g., axis areas) of the flexible screen

over a period of time seeking an indication of at least one of bending, rolling, and/or curving of the flexible display screen. In one embodiment, as described with reference to FIG. 2A, non-touch logic **205** may be further used to facilitate non-touch sensor(s) **233** (e.g., current delta sensors) to measure any changes in the current charges in and around the one or more specific areas of the flexible screen that indicates, for example, bending of the flexible screen, where this measuring includes detecting differences in charges by comparing one or more present current charges with one or more previous current charges over a period of time.

(56) Referring back to method **300**, in one embodiment, at block **307**, the one or more areas are identified as zones and, at block **309**, the positions of flexible display screen within the zones are marked as segments. At block **311**, segments are activated, including associating user interfaces to the segments. At block **313**, display of contents via the segments is allowed based on user preferences, etc.

(57) FIG. 4 illustrates an embodiment of a computing system **400** capable of supporting the operations discussed above. Computing system **400** represents a range of computing and electronic devices (wired or wireless) including, for example, desktop computing systems, laptop computing systems, cellular telephones, personal digital assistants (PDAs) including cellular-enabled PDAs, set top boxes, smartphones, tablets, wearable devices, etc. Alternate computing systems may include more, fewer and/or different components. Computing device **400** may be the same as or similar to or include computing devices **100** described in reference to FIG. 1.

(58) Computing system **400** includes bus **405** (or, for example, a link, an interconnect, or another type of communication device or interface to communicate information) and processor **410** coupled to bus **405** that may process information. While computing system **400** is illustrated with a single processor, it may include multiple processors and/or co-processors, such as one or more of central processors, image signal processors, graphics processors, and vision processors, etc. Computing system **400** may further include random access memory (RAM) or other dynamic storage device **420** (referred to as main memory), coupled to bus **405** and may store information and instructions that may be executed by processor **410**. Main memory **420** may also be used to store temporary variables or other intermediate information during execution of instructions by processor **410**.

(59) Computing system **400** may also include read only memory (ROM) and/or other storage device **430** coupled to bus **405** that may store static information and instructions for processor **410**. Data storage device **440** may be coupled to bus **405** to store information and instructions. Data storage device **440**, such as magnetic disk or optical disc and corresponding drive may be coupled to computing system **400**.

(60) Computing system **400** may also be coupled via bus **405** to display device **450**, such as a cathode ray tube (CRT), liquid crystal display (LCD) or Organic Light Emitting Diode (OLED) array, to display information to a user. User input device **460**, including alphanumeric and other keys, may be coupled to bus **405** to communicate information and command selections to processor **410**. Another type of user input device **460** is cursor control **470**, such as a mouse, a trackball, a touchscreen, a touchpad, or cursor direction keys to communicate direction information and command selections to processor **410** and to control cursor movement on display **450**. Camera and microphone arrays **490** of computer system **400** may be coupled to bus **405** to observe gestures, record audio and video and to receive and transmit visual and audio commands.

(61) Computing system **400** may further include network interface(s) **480** to provide access to a network, such as a local area network (LAN), a wide area network (WAN), a metropolitan area network (MAN), a personal area network (PAN), Bluetooth, a cloud network, a mobile network (e.g., 3.sup.rd Generation (3G), etc.), an intranet, the Internet, etc. Network interface(s) **480** may include, for example, a wireless network interface having antenna **485**, which may represent one or more antenna(c). Network interface(s) **480** may also include, for example, a wired network interface to communicate with remote devices via network cable **487**, which may be, for example, an Ethernet cable, a coaxial cable, a fiber optic cable, a serial cable, or a parallel cable.

- (62) Network interface(s) **480** may provide access to a LAN, for example, by conforming to IEEE 802.11b and/or IEEE 802.11g standards, and/or the wireless network interface may provide access to a personal area network, for example, by conforming to Bluetooth standards. Other wireless network interfaces and/or protocols, including previous and subsequent versions of the standards, may also be supported.
- (63) In addition to, or instead of, communication via the wireless LAN standards, network interface(s) **480** may provide wireless communication using, for example, Time Division, Multiple Access (TDMA) protocols, Global Systems for Mobile Communications (GSM) protocols, Code Division, Multiple Access (CDMA) protocols, and/or any other type of wireless communications protocols.
- (64) Network interface(s) **480** may include one or more communication interfaces, such as a modem, a network interface card, or other well-known interface devices, such as those used for coupling to the Ethernet, token ring, or other types of physical wired or wireless attachments for purposes of providing a communication link to support a LAN or a WAN, for example. In this manner, the computer system may also be coupled to a number of peripheral devices, clients, control surfaces, consoles, or servers via a conventional network infrastructure, including an Intranet or the Internet, for example.
- (65) It is to be appreciated that a lesser or more equipped system than the example described above may be preferred for certain implementations. Therefore, the configuration of computing system **400** may vary from implementation to implementation depending upon numerous factors, such as price constraints, performance requirements, technological improvements, or other circumstances. Examples of the electronic device or computer system **400** may include without limitation a mobile device, a personal digital assistant, a mobile computing device, a smartphone, a cellular telephone, a handset, a one-way pager, a two-way pager, a messaging device, a computer, a personal computer (PC), a desktop computer, a laptop computer, a notebook computer, a handheld computer, a tablet computer, a server, a server array or server farm, a web server, a network server, an Internet server, a work station, a mini-computer, a main frame computer, a supercomputer, a network appliance, a web appliance, a distributed computing system, multiprocessor systems, processor-based systems, consumer electronics, programmable consumer electronics, television, digital television, set top box, wireless access point, base station, subscriber station, mobile subscriber center, radio network controller, router, hub, gateway, bridge, switch, machine, or combinations thereof.
- (66) Embodiments may be implemented as any or a combination of: one or more microchips or integrated circuits interconnected using a parentboard, hardwired logic, software stored by a memory device and executed by a microprocessor, firmware, an application specific integrated circuit (ASIC), and/or a field programmable gate array (FPGA). The term “logic” may include, by way of example, software or hardware and/or combinations of software and hardware.
- (67) Embodiments may be provided, for example, as a computer program product which may include one or more machine-readable media having stored thereon machine-executable instructions that, when executed by one or more machines such as a computer, network of computers, or other electronic devices, may result in the one or more machines carrying out operations in accordance with embodiments described herein. A machine-readable medium may include, but is not limited to, floppy diskettes, optical disks, CD-ROMs (Compact Disc-Read Only Memories), and magneto-optical disks, ROMs, RAMs, EPROMs (Erasable Programmable Read Only Memories), EEPROMs (Electrically Erasable Programmable Read Only Memories), magnetic or optical cards, flash memory, or other type of media/machine-readable medium suitable for storing machine-executable instructions.
- (68) Moreover, embodiments may be downloaded as a computer program product, wherein the program may be transferred from a remote computer (e.g., a server) to a requesting computer (e.g., a client) by way of one or more data signals embodied in and/or modulated by a carrier wave or other propagation medium via a communication link (e.g., a modem and/or network connection).

(69) References to “one embodiment”, “an embodiment”, “example embodiment”, “various embodiments”, etc., indicate that the embodiment(s) so described may include particular features, structures, or characteristics, but not every embodiment necessarily includes the particular features, structures, or characteristics. Further, some embodiments may have some, all, or none of the features described for other embodiments.

(70) In the following description and claims, the term “coupled” along with its derivatives, may be used. “Coupled” is used to indicate that two or more elements co-operate or interact with each other, but they may or may not have intervening physical or electrical components between them.

(71) As used in the claims, unless otherwise specified the use of the ordinal adjectives “first”, “second”, “third”, etc., to describe a common element, merely indicate that different instances of like elements are being referred to, and are not intended to imply that the elements so described must be in a given sequence, either temporally, spatially, in ranking, or in any other manner.

(72) FIG. 5 illustrates an embodiment of a computing environment **500** capable of supporting the operations discussed above. The modules and systems can be implemented in a variety of different hardware architectures and form factors including that shown in FIG. 9.

(73) The Command Execution Module **501** includes a central processing unit to cache and execute commands and to distribute tasks among the other modules and systems shown. It may include an instruction stack, a cache memory to store intermediate and final results, and mass memory to store applications and operating systems. The Command Execution Module may also serve as a central coordination and task allocation unit for the system.

(74) The Screen Rendering Module **521** draws objects on the one or more multiple screens for the user to see. It can be adapted to receive the data from the Virtual Object Behavior Module **504**, described below, and to render the virtual object and any other objects and forces on the appropriate screen or screens. Thus, the data from the Virtual Object Behavior Module would determine the position and dynamics of the virtual object and associated gestures, forces and objects, for example, and the Screen Rendering Module would depict the virtual object and associated objects and environment on a screen, accordingly. The Screen Rendering Module could further be adapted to receive data from the Adjacent Screen Perspective Module **507**, described below, to either depict a target landing area for the virtual object if the virtual object could be moved to the display of the device with which the Adjacent Screen Perspective Module is associated. Thus, for example, if the virtual object is being moved from a main screen to an auxiliary screen, the Adjacent Screen Perspective Module 2 could send data to the Screen Rendering Module to suggest, for example in shadow form, one or more target landing areas for the virtual object on that track to a user's hand movements or eye movements.

(75) The Object and Gesture Recognition System **522** may be adapted to recognize and track hand and arm gestures of a user. Such a module may be used to recognize hands, fingers, finger gestures, hand movements and a location of hands relative to displays. For example, the Object and Gesture Recognition Module could for example determine that a user made a body part gesture to drop or throw a virtual object onto one or the other of the multiple screens, or that the user made a body part gesture to move the virtual object to a bezel of one or the other of the multiple screens. The Object and Gesture Recognition System may be coupled to a camera or camera array, a microphone or microphone array, a touch screen or touch surface, or a pointing device, or some combination of these items, to detect gestures and commands from the user.

(76) The touch screen or touch surface of the Object and Gesture Recognition System may include a touch screen sensor. Data from the sensor may be fed to hardware, software, firmware or a combination of the same to map the touch gesture of a user's hand on the screen or surface to a corresponding dynamic behavior of a virtual object. The sensor data may be used to momentum and inertia factors to allow a variety of momentum behavior for a virtual object based on input from the user's hand, such as a swipe rate of a user's finger relative to the screen. Pinching gestures may be interpreted as a command to lift a virtual object from the display screen, or to begin

generating a virtual binding associated with the virtual object or to zoom in or out on a display. Similar commands may be generated by the Object and Gesture Recognition System using one or more cameras without benefit of a touch surface.

(77) The Direction of Attention Module **523** may be equipped with cameras or other sensors to track the position or orientation of a user's face or hands. When a gesture or voice command is issued, the system can determine the appropriate screen for the gesture. In one example, a camera is mounted near each display to detect whether the user is facing that display. If so, then the direction of attention module information is provided to the Object and Gesture Recognition Module **522** to ensure that the gestures or commands are associated with the appropriate library for the active display. Similarly, if the user is looking away from all of the screens, then commands can be ignored.

(78) The Device Proximity Detection Module **525** can use proximity sensors, compasses, GPS (global positioning system) receivers, personal area network radios, and other types of sensors, together with triangulation and other techniques to determine the proximity of other devices. Once a nearby device is detected, it can be registered to the system and its type can be determined as an input device or a display device or both. For an input device, received data may then be applied to the Object Gesture and Recognition System **522**. For a display device, it may be considered by the Adjacent Screen Perspective Module **507**.

(79) The Virtual Object Behavior Module **504** is adapted to receive input from the Object Velocity and Direction Module, and to apply such input to a virtual object being shown in the display. Thus, for example, the Object and Gesture Recognition System would interpret a user gesture and by mapping the captured movements of a user's hand to recognized movements, the Virtual Object Tracker Module would associate the virtual object's position and movements to the movements as recognized by Object and Gesture Recognition System, the Object and Velocity and Direction Module would capture the dynamics of the virtual object's movements, and the Virtual Object Behavior Module would receive the input from the Object and Velocity and Direction Module to generate data that would direct the movements of the virtual object to correspond to the input from the Object and Velocity and Direction Module.

(80) The Virtual Object Tracker Module **506** on the other hand may be adapted to track where a virtual object should be located in three dimensional space in a vicinity of an display, and which body part of the user is holding the virtual object, based on input from the Object and Gesture Recognition Module. The Virtual Object Tracker Module **506** may for example track a virtual object as it moves across and between screens and track which body part of the user is holding that virtual object. Tracking the body part that is holding the virtual object allows a continuous awareness of the body part's air movements, and thus an eventual awareness as to whether the virtual object has been released onto one or more screens.

(81) The Gesture to View and Screen Synchronization Module **508**, receives the selection of the view and screen or both from the Direction of Attention Module **523** and, in some cases, voice commands to determine which view is the active view and which screen is the active screen. It then causes the relevant gesture library to be loaded for the Object and Gesture Recognition System **522**. Various views of an application on one or more screens can be associated with alternative gesture libraries or a set of gesture templates for a given view. As an example in FIG. **1A** a pinch-release gesture launches a torpedo, but in FIG. **1B**, the same gesture launches a depth charge.

(82) The Adjacent Screen Perspective Module **507**, which may include or be coupled to the Device Proximity Detection Module **525**, may be adapted to determine an angle and position of one display relative to another display. A projected display includes, for example, an image projected onto a wall or screen. The ability to detect a proximity of a nearby screen and a corresponding angle or orientation of a display projected therefrom may for example be accomplished with either an infrared emitter and receiver, or electromagnetic or photo-detection sensing capability. For technologies that allow projected displays with touch input, the incoming video can be analyzed to

determine the position of a projected display and to correct for the distortion caused by displaying at an angle. An accelerometer, magnetometer, compass, or camera can be used to determine the angle at which a device is being held while infrared emitters and cameras could allow the orientation of the screen device to be determined in relation to the sensors on an adjacent device. The Adjacent Screen Perspective Module **507** may, in this way, determine coordinates of an adjacent screen relative to its own screen coordinates. Thus, the Adjacent Screen Perspective Module may determine which devices are in proximity to each other, and further potential targets for moving one or more virtual object's across screens. The Adjacent Screen Perspective Module may further allow the position of the screens to be correlated to a model of three-dimensional space representing all of the existing objects and virtual objects.

(83) The Object and Velocity and Direction Module **503** may be adapted to estimate the dynamics of a virtual object being moved, such as its trajectory, velocity (whether linear or angular), momentum (whether linear or angular), etc. by receiving input from the Virtual Object Tracker Module. The Object and Velocity and Direction Module may further be adapted to estimate dynamics of any physics forces, by for example estimating the acceleration, deflection, degree of stretching of a virtual binding, etc. and the dynamic behavior of a virtual object once released by a user's body part. The Object and Velocity and Direction Module may also use image motion, size and angle changes to estimate the velocity of objects, such as the velocity of hands and fingers

(84) The Momentum and Inertia Module **502** can use image motion, image size, and angle changes of objects in the image plane or in a three-dimensional space to estimate the velocity and direction of objects in the space or on a display. The Momentum and Inertia Module is coupled to the Object and Gesture Recognition System **522** to estimate the velocity of gestures performed by hands, fingers, and other body parts and then to apply those estimates to determine momentum and velocities to virtual objects that are to be affected by the gesture.

(85) The 3D Image Interaction and Effects Module **505** tracks user interaction with 3D images that appear to extend out of one or more screens. The influence of objects in the z-axis (towards and away from the plane of the screen) can be calculated together with the relative influence of these objects upon each other. For example, an object thrown by a user gesture can be influenced by 3D objects in the foreground before the virtual object arrives at the plane of the screen. These objects may change the direction or velocity of the projectile or destroy it entirely. The object can be rendered by the 3D Image Interaction and Effects Module in the foreground on one or more of the displays.

(86) The following clauses and/or examples pertain to further embodiments or examples. Specifics in the examples may be used anywhere in one or more embodiments. The various features of the different embodiments or examples may be variously combined with some features included and others excluded to suit a variety of different applications. Examples may include subject matter such as a method, means for performing acts of the method, at least one machine-readable medium including instructions that, when performed by a machine cause the machine to performs acts of the method, or of an apparatus or system for facilitating hybrid communication according to embodiments and examples described herein.

(87) Some embodiments pertain to Example 1 that includes an apparatus to facilitate dynamic detection and intelligent use of segmentation on flexible display screens, comprising: a flexible display screen; touch segment detection logic to detect, via one or more touch sensors, alterations in current in and around one or more areas of the flexible display screen, wherein the alterations represent pressure being applied to cause at least one of bending, rolling, and curving of the flexible display screen at the one or more areas; marking/dividing logic to divide the flexible display screen into a plurality of zones corresponding to the one or more areas, wherein the marking/dividing logic is further to mark a plurality of portions of the plurality of zones to serve as a plurality of segments; and content/use logic to facilitate displaying of contents via the plurality of segments of the flexible display screen.

(88) Example 2 includes the subject matter of Example 1, further comprising non-touch segment detection logic to detect, via one or more non-touch sensors, current charges, over a period of time, in and around the one or more areas of the flexible display screen, wherein the non-touch segment detection logic is further to measure gradual changes in the current charges over the period of time by detecting and comparing one or more present current charges with one or more previous current charges, wherein the gradual changes represent the applied pressure.

(89) Example 3 includes the subject matter of Example 1 or 2, further comprising application/user interface logic to associate a plurality of user interface to the plurality of segments, wherein a user interface is associated with each of the plurality of segments, wherein the application/user interface logic is further to facilitate interactivity amongst the plurality of segments.

(90) Example 4 includes the subject matter of Example 1, wherein each segment of the plurality of segments serves as a segment display screen on the flexible display screen, wherein the segment display screen is distinct from other segments of the plurality of segments serving as segment display screens on the flexible display screen.

(91) Example 5 includes the subject matter of Example 1, wherein the contents comprise a plurality of content portions, wherein each segment of the plurality of segments to display one or more content portions of the plurality of content portions, wherein the one or more content portions are different from other content portions of the plurality of content portions displayed via other segments of the plurality of segments.

(92) Example 6 includes the subject matter of Example 1 or 2, further comprising preferences logic to facilitate maintenance of user preferences provided by a user associated with the flexible display screen.

(93) Example 7 includes the subject matter of Example 1, wherein the contents are displayed via the plurality of segments based on the user preferences, wherein the user preferences include personal settings relating to one or more of display adjustments, content selections, and interactivity settings.

(94) Example 8 includes the subject matter of Example 1, wherein the flexible display screen comprises at least one of a standalone flexible display screen and a device-based flexible display screen mounted on a computing device including at least one of a wearable device, smart window, smart mobile device, laptop computer, desktop computer, and server computer, wherein the device-based flexible display screen includes an extension screen of a main display screen of the computing device.

(95) Some embodiments pertain to Example 9 that includes a method for facilitating dynamic detection and intelligent use of segmentation on flexible display screens, comprising: detecting, via one or more touch sensors, alterations in current in and around one or more areas of a flexible display screen, wherein the alterations represent pressure being applied to cause at least one of bending, rolling, and curving of the flexible display screen at the one or more areas; dividing the flexible display screen into a plurality of zones corresponding to the one or more areas, wherein dividing further includes marking a plurality of portions of the plurality of zones to serve as a plurality of segments; and facilitating displaying of contents via the plurality of segments of the flexible display screen.

(96) Example 10 includes the subject matter of Example 9, further comprising: detecting, via one or more non-touch sensors, current charges, over a period of time, in and around the one or more areas of the flexible display screen; and measuring gradual changes in the current charges over the period of time by detecting and comparing one or more present current charges with one or more previous current charges, wherein the gradual changes represent the applied pressure.

(97) Example 11 includes the subject matter of Example 9 or 10, further comprising:

(98) associating a plurality of user interface to the plurality of segments, wherein a user interface is associated with each of the plurality of segments; and facilitating interactivity amongst the plurality of segments.

- (99) Example 12 includes the subject matter of Example 9, wherein each segment of the plurality of segments serves as a segment display screen on the flexible display screen, wherein the segment display screen is distinct from other segments of the plurality of segments serving as segment display screens on the flexible display screen.
- (100) Example 13 includes the subject matter of Example 9, wherein the contents comprise a plurality of content portions, wherein each segment of the plurality of segments to display one or more content portions of the plurality of content portions, wherein the one or more content portions are different from other content portions of the plurality of content portions displayed via other segments of the plurality of segments.
- (101) Example 14 includes the subject matter of Example 9 or 10, further comprising facilitating maintenance of user preferences provided by a user associated with the flexible display screen.
- (102) Example 15 includes the subject matter of Example 9, wherein the contents are displayed via the plurality of segments based on the user preferences, wherein the user preferences include personal settings relating to one or more of display adjustments, content selections, and interactivity settings.
- (103) Example 16 includes the subject matter of Example 9, wherein the flexible display screen comprises at least one of a standalone flexible display screen and a device-based flexible display screen mounted on a computing device including at least one of a wearable device, smart window, smart mobile device, laptop computer, desktop computer, and server computer, wherein the device-based flexible display screen includes an extension screen of a main display screen of the computing device.
- (104) Example 17 includes at least one machine-readable medium comprising a plurality of instructions, when executed on a computing device, to implement or perform a method or realize an apparatus as claimed in any preceding examples, embodiments, or claims.
- (105) Example 18 includes at least one non-transitory or tangible machine-readable medium comprising a plurality of instructions, when executed on a computing device, to implement or perform a method or realize an apparatus as claimed in any preceding examples, embodiments, or claims.
- (106) Example 19 includes a system comprising a mechanism to implement or perform a method or realize an apparatus as claimed in any preceding examples, embodiments, or claims.
- (107) Example 20 includes an apparatus comprising means to perform a method as claimed in any preceding examples, embodiments, or claims.
- (108) Example 21 includes a computing device arranged to implement or perform a method or realize an apparatus as claimed in any preceding examples, embodiments, or claims.
- (109) Example 22 includes a communications device arranged to implement or perform a method or realize an apparatus as claimed in any preceding examples, embodiments, or claims.
- (110) Some embodiments pertain to Example 23 includes a system comprising a storage device having instructions, and a processor to execute the instructions to facilitate a mechanism to perform one or more operations comprising: detecting, via one or more touch sensors, alterations in current in and around one or more areas of a flexible display screen, wherein the alterations represent pressure being applied to cause at least one of bending, rolling, and curving of the flexible display screen at the one or more areas; dividing the flexible display screen into a plurality of zones corresponding to the one or more areas, wherein dividing further includes marking a plurality of portions of the plurality of zones to serve as a plurality of segments; and facilitating displaying of contents via the plurality of segments of the flexible display screen.
- (111) Example 24 includes the subject matter of Example 23, wherein the one or more operations further comprise: detecting, via one or more non-touch sensors, current charges, over a period of time, in and around the one or more areas of the flexible display screen; and measuring gradual changes in the current charges over the period of time by detecting and comparing one or more present current charges with one or more previous current charges, wherein the gradual changes

represent the applied pressure.

(112) Example 25 includes the subject matter of Example 23 or 24, wherein the one or more operations further comprise: associating a plurality of user interface to the plurality of segments, wherein a user interface is associated with each of the plurality of segments; and facilitating interactivity amongst the plurality of segments.

(113) Example 26 includes the subject matter of Example 23, wherein each segment of the plurality of segments serves as a segment display screen on the flexible display screen, wherein the segment display screen is distinct from other segments of the plurality of segments serving as segment display screens on the flexible display screen.

(114) Example 27 includes the subject matter of Example 23, wherein the contents comprise a plurality of content portions, wherein each segment of the plurality of segments to display one or more content portions of the plurality of content portions, wherein the one or more content portions are different from other content portions of the plurality of content portions displayed via other segments of the plurality of segments.

(115) Example 28 includes the subject matter of Example 23 or 24, wherein the one or more operations further comprise facilitating maintenance of user preferences provided by a user associated with the flexible display screen.

(116) Example 29 includes the subject matter of Example 23, wherein the contents are displayed via the plurality of segments based on the user preferences, wherein the user preferences include personal settings relating to one or more of display adjustments, content selections, and interactivity settings.

(117) Example 30 includes the subject matter of Example 23, wherein the flexible display screen comprises at least one of a standalone flexible display screen and a device-based flexible display screen mounted on a computing device including at least one of a wearable device, smart window, smart mobile device, laptop computer, desktop computer, and server computer, wherein the device-based flexible display screen includes an extension screen of a main display screen of the computing device.

(118) Some embodiments pertain to Example 31 includes an apparatus comprising: means for detecting, via one or more touch sensors, alterations in current in and around one or more areas of a flexible display screen, wherein the alterations represent pressure being applied to cause at least one of bending, rolling, and curving of the flexible display screen at the one or more areas; means for dividing the flexible display screen into a plurality of zones corresponding to the one or more areas, wherein dividing further includes marking a plurality of portions of the plurality of zones to serve as a plurality of segments; and means for facilitating displaying of contents via the plurality of segments of the flexible display screen.

(119) Example 32 includes the subject matter of Example 31, wherein the one or more operations further comprise: means for detecting, via one or more non-touch sensors, current charges, over a period of time, in and around the one or more areas of the flexible display screen; and means for measuring gradual changes in the current charges over the period of time by detecting and comparing one or more present current charges with one or more previous current charges, wherein the gradual changes represent the applied pressure.

(120) Example 33 includes the subject matter of Example 31 or 32, further comprising: means for associating a plurality of user interface to the plurality of segments, wherein a user interface is associated with each of the plurality of segments; and means for facilitating interactivity amongst the plurality of segments.

(121) Example 34 includes the subject matter of Example 31, wherein each segment of the plurality of segments serves as a segment display screen on the flexible display screen, wherein the segment display screen is distinct from other segments of the plurality of segments serving as segment display screens on the flexible display screen.

(122) Example 35 includes the subject matter of Example 31, wherein the contents comprise a

plurality of content portions, wherein each segment of the plurality of segments to display one or more content portions of the plurality of content portions, wherein the one or more content portions are different from other content portions of the plurality of content portions displayed via other segments of the plurality of segments.

(123) Example 36 includes the subject matter of Example 31 or 32, further comprising means for facilitating maintenance of user preferences provided by a user associated with the flexible display screen.

(124) Example 37 includes the subject matter of Example 31, wherein the contents are displayed via the plurality of segments based on the user preferences, wherein the user preferences include personal settings relating to one or more of display adjustments, content selections, and interactivity settings.

(125) Example 38 includes the subject matter of Example 31, wherein the flexible display screen comprises at least one of a standalone flexible display screen and a device-based flexible display screen mounted on a computing device including at least one of a wearable device, smart window, smart mobile device, laptop computer, desktop computer, and server computer, wherein the device-based flexible display screen includes an extension screen of a main display screen of the computing device.

(126) Example 39 includes at least one non-transitory or tangible machine-readable medium comprising a plurality of instructions, when executed on a computing device, to implement or perform a method as claimed in any of examples, embodiments, or claims 9-16.

(127) Example 40 includes at least one machine-readable medium comprising a plurality of instructions, when executed on a computing device, to implement or perform a method as claimed in any of examples, embodiments, or claims 9-16.

(128) Example 41 includes a system comprising a mechanism to implement or perform a method as claimed in any of examples, embodiments, or claims 9-16.

(129) Example 42 includes an apparatus comprising means for performing a method as claimed in any of examples, embodiments, or claims 9-16.

(130) Example 43 includes a computing device arranged to implement or perform a method as claimed in any of examples, embodiments, or claims 9-16.

(131) Example 44 includes a communications device arranged to implement or perform a method as claimed in any of examples, embodiments, or claims 9-16.

(132) The drawings and the forgoing description give examples of embodiments. Those skilled in the art will appreciate that one or more of the described elements may well be combined into a single functional element. Alternatively, certain elements may be split into multiple functional elements. Elements from one embodiment may be added to another embodiment. For example, orders of processes described herein may be changed and are not limited to the manner described herein. Moreover, the actions any flow diagram need not be implemented in the order shown; nor do all of the acts necessarily need to be performed. Also, those acts that are not dependent on other acts may be performed in parallel with the other acts. The scope of embodiments is by no means limited by these specific examples. Numerous variations, whether explicitly given in the specification or not, such as differences in structure, dimension, and use of material, are possible. The scope of embodiments is at least as broad as given by the following claims.

Claims

1. A portable computing device comprising: a flexible touchscreen having a flat state and a bent state; one or more storage devices; and instructions; and processor circuitry to execute the instructions to at least: cause display of a first user interface of an application in a first segment on a first side of a bend of the flexible touchscreen after detection of a change from the flat state to the bent state, and cause display of a second user interface of the application and a control interface in

a second segment on a second side of the bend of the flexible touchscreen, the second user interface different than the first user interface.

2. The portable computing device of claim 1, wherein the application is a game, the first user interface includes content of the game.

3. The portable computing device as defined in claim 1, wherein the first segment and the second segment are separated along an axis of the bend.

4. The portable computing device as defined in claim 1, further including a sensor to detect a change in the flexible touchscreen.

5. The portable computing device as defined in claim 4, wherein the processor circuitry is to cause display of the first user interface responsive to an output from the sensor.

6. The portable computing device as defined in claim 4, wherein the flexible touchscreen includes the sensor.

7. The portable computing device as defined in claim 1, wherein the portable computing device operates as a phone in a flat state.

8. The portable computing device as defined in claim 1, wherein the bent state includes a curve.

9. A portable computing device comprising: a flexible touchscreen having an unfolded state and a folded state; one or more storage devices; and instructions; and processor circuitry to execute the instructions to at least: cause display of content of a game in a first segment of the flexible touchscreen, the first segment on a first side of a bend of the flexible touchscreen when the flexible touchscreen is in the folded state, and cause display of a control interface and a control interface of the game in a second segment of the flexible touchscreen on a second side of the bend of the flexible touchscreen.

10. The portable computing device of claim 9, wherein the control interface includes controls to play the game.

11. The portable computing device of claim 9, wherein the control interface includes controls for a user to select elements for use within the game.

12. The portable computing device of claim 9, wherein the first segment and the second segment are separated along an axis of the bend.

13. The portable computing device of claim 9, further including a sensor to detect a change in the flexible touchscreen.

14. The portable computing device of claim 9, wherein the portable computing device is a laptop computer.

15. A non-transitory computer readable medium comprising first instructions that, when executed, cause at least one first programmable circuit to at least: cause transmission of second instructions, the second instructions, when executed, to cause at least one second programmable circuit to at least: cause display of a first user interface of an application in a first segment on a first side of a bend of a flexible touchscreen after detection of a change from a flat state to a bent state of the flexible touchscreen, and cause display of a second user interface and a control interface of the application in a second segment on a second side of the bend of the flexible touchscreen, the second user interface different than the first user interface.

16. The non-transitory computer readable medium of claim 15, wherein the application is a game, the first user interface includes content of the game.

17. The non-transitory computer readable medium as defined in claim 15, wherein the first segment and the second segment are separated along an axis of the bend.

18. The non-transitory computer readable medium as defined in claim 15, wherein the second instructions, when executed, cause the at least one second programmable circuit to cause display of the first user interface responsive to an output from a sensor to detect the bend.

19. The non-transitory computer readable medium as defined in claim 15, wherein the first instructions are to cause transmission of the second instructions to the at least one second programmable circuit.

20. The non-transitory computer readable medium as defined in claim 15, wherein the second instructions include the application.

21. A mobile communication device comprising: an antenna; a touchscreen display including a first display segment, a second display segment, and a third display segment, wherein the first display segment is separated from the second display segment along a first axis, and wherein the second display segment is separated from the third display segment along a second axis, the touchscreen display movable between a first state and a second state; communication circuitry to wirelessly communicate with another device via the antenna; one or more storage devices; and processor circuitry to execute instructions to: display a first portion of a first application on the first display segment of the touchscreen display in response to detecting a transition of the touchscreen display from the first state to the second state, the first display segment including a first user interface associated with the first application; display a second portion of the first application and a control interface on the second display segment of the touchscreen display, the second display segment including a second user interface associated with the first application; and display a third portion of the first application on the third display segment of the touchscreen display, the third display segment including a third user interface associated with the first application.

22. The mobile communication device as defined in claim 21, wherein the first axis is along a bend in the touchscreen display.

23. The mobile communication device as defined in claim 21, further including a sensor to detect a change in a state of the touchscreen display.

24. The mobile communication device as defined in claim 23, wherein the processor circuitry is to cause display of the first portion responsive to an output from the sensor.

25. The mobile communication device as defined in claim 21, processor circuitry to execute instructions to display a fourth portion of the first application on a fourth display segment of the touchscreen display.
